

claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-ad3c2f541432358e6.jsonl`  
- SHA-256: `f5ab88f50f1c4cd987c4a79f82c4ec3e0167ab84034338085c81aa45070f6b7e`  
- Source modified: `2026-06-06T17:30:48+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_6237b110-d58`  
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`
```

Transcript

1. user (2026-06-06T17:28:43.708Z)

Web Searcher: contrarian/distillation-vs-weak

Research question: "Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):

- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:

1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ? especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?
5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio)."

Your angle: ****contrarian/distillation-vs-weak**** ? Adversarially tests the user's current Gemini-teacher ? local-student plan against weak/self-supervised alternatives, surfacing the few-labels overfitting limits they are already hitting at F1?0.44.
Search query: `VLM teacher student distillation video temporal segmentation vs weakly supervised self-supervised action localization limited labels`

Task

Use WebSearch with the query above (or a refined version). Return the top 4-6 most relevant results.

Rank by relevance to the ORIGINAL question, not just the search query. Skip obvious SEO spam/content farms.

Include a short snippet capturing why each result is relevant.

Structured output only.

2. assistant (2026-06-06T17:28:47.509Z)

I'll research this question. Let me start with the suggested search query and refine from there.

3. user (2026-06-06T17:28:59.132Z)

Web search results for query: "badminton rally segmentation audio shuttle hit detection ShuttleSet dataset license temporal action localization"

Links: [{"title": "[2306.04948] ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis", "url": "https://arxiv.org/html/2306.04948"}, {"title": "ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis | Request PDF", "url": "https://www.researchgate.net/publication/372910678_ShuttleSet_A_Human-Annotated_Stroke-Level_Singles_Dataset_for_Badminton_Tactical_Analysis"}, {"title": "[2306.04948] ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis", "url": "https://arxiv.org/abs/2306.04948"}, {"title": "(PDF) ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis", "url": "https://www.researchgate.net/publication/371413847_ShuttleSet_A_Human-Annotated_Stroke-Level_Singles_Dataset_for_Badminton_Tactical_Analysis"}, {"title": "MonoTrack: Shuttle trajectory reconstruction from monocular badminton video", "url": "https://arxiv.org/pdf/2204.01899"}, {"title": "ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis | Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining", "url": "https://dl.acm.org/doi/10.1145/3580305.3599906"}, {"title": "VideoBadminton: A Video Dataset for Badminton Action Recognition | Request PDF", "url": "https://www.researchgate.net/publication/388099044_VideoBadminton_A_Video_Dataset_for_Badminton_Action_Recognition"}, {"title": "BST: Badminton Stroke-type Transformer for Skeleton-based Action Recognition in Racket Sports", "url": "https://arxiv.org/pdf/2502.21085"}, {"title": "BFMD: A Full-Match Badminton Dense Dataset for Dense Shot Captioning", "url": "https://arxiv.org/pdf/2603.25533"}]

Based on the search results, here's what I found about badminton rally segmentation, shuttle hit detection, and the ShuttleSet dataset:

ShuttleSet Dataset Overview

ShuttleSet is the largest publicly-available badminton singles dataset with annotated stroke-level records, containing 104 sets, 3,685 rallies, and 36,492 strokes in 44 matches between 2018 and 2021 with 27 top-ranking men's singles and women's singles players.

Rally Segmentation and Hit Detection

Identifying the shots of a rally is critical for reconstructing 3D trajectories and other downstream applications. A shot happens when a player hits the shuttle with her racket and ends right before the opposing player hits the shuttle or if the shuttle hits the court. Therefore, in order to segment the video into successive shots, it is equivalent to identify the hit events.

Hit detection is formulated as a multi-class classification problem, with three-way classification with labels no hit, near player hit, and far player hit predicted at each frame.

Dataset Annotation

ShuttleSet is manually annotated with a computer-aided labeling tool with a choice of 18 distinct shot type classes, the corresponding hitting locations, and the locations of both players.

Related Dataset

There is also the Badminton Full Match Dense (BFMD) dataset, with 19 broadcast matches covering over 20 hours of play, comprising 1,687 rallies and 16,751 hit events, each annotated with a shot caption.

Regarding the specific aspects of dataset license and temporal action localization, those details were not found in the search results provided.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

4. user (2026-06-06T17:29:04.390Z)

Web search results for query: "VLM teacher student distillation video temporal segmentation vs weakly supervised self-supervised action localization limited labels"

Links: [{"title": "Weakly Supervised Temporal Action Localization with Segment-Level Labels", "url": "https://arxiv.org/pdf/2007.01598"}, {"title": "Improving Weakly Supervised Temporal Action Localization by Exploiting Multi-resolution Information in Temporal Domain", "url": "https://arxiv.org/pdf/2506.18261"}, {"title": "Disentangling Spatio-temporal Knowledge for Weakly Supervised Object", "url": "https://openaccess.thecvf.com/content/WACV2025/papers/Liao_Disentangling_Spatio-Temporal_Knowledge_for_Weakly_Supervised_Object_Detection_and_Segmentation_WACV_2025_paper.pdf"}, {"title": "Probabilistic Vision-Language Representation for Weakly Supervised Temporal Action Localization", "url": "https://arxiv.org/pdf/2408.05955"}, {"title": "Weakly Supervised Temporal Action Localization Using Deep Metric Learning", "url": "https://arxiv.org/pdf/2001.07793"}, {"title": "ASM-Loc: Action-aware Segment Modeling for Weakly-Supervised Temporal Action Localization", "url": "https://arxiv.org/pdf/2203.15187"}, {"title": "Distilling Vision-Language Pre-training to Collaborate with Weakly-Supervised Temporal Action Localization", "url": "https://arxiv.org/pdf/2212.09335"}, {"title": "Weakly supervised temporal action localization via a multimodal feature map diffusion process - ScienceDirect", "url": "https://www.sciencedirect.com/science/article/abs/pii/S0952197625010449"}]

I found relevant research on your query topic. Here are the key findings:

Overview of the Field

Weakly supervised temporal action localization (WTAL) is proposed to alleviate the laborious annotation procedure for Temporal Action Localization, training with only video-level labels. This directly addresses the "limited labels" aspect of your query.

Vision-Language Models (VLMs) and Distillation

Recent research makes exploration to steer VLP (Vision-Language Pre-training) for long video temporal localization, under the weakly-supervised setting. More specifically, a Distillation-Collaboration Framework covers two parallel branches, named CBP and VLP, and distill confident background pseudo-labels from the CBP branch, to train prompt vectors and temporal Transformer in the VLP branch, while confident foreground pseudo-labels are distilled for the CBP branch.

Knowledge Distillation Approaches

Knowledge distillation approaches deploy distillation to disentangle spatial and temporal information for weakly supervised learning with video-level labels, where knowledge in activation map is transferred from teacher to student.

Challenges in Temporal Segmentation

Due to the absence of frame-level annotations, the temporal class activation map under weak supervision suffers from poor foreground-background separability, leading to inaccurate predictions of action boundaries, often resulting in over-complete or incomplete action localization.

The research shows an active area combining VLMs, teacher-student distillation, and weakly-supervised approaches to address temporal action localization with limited annotations.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

5. assistant (2026-06-06T17:29:08.082Z)

The first query surfaced the key contrarian paper (Distilling VLP to Collaborate with WTAL). Let me dig into more angles: audio-based hit detection specifically, and the few-labels/overfitting limits of distillation.

6. user (2026-06-06T17:29:17.844Z)

Web search results for query: "audio onset detection racket sport hit ball impact sound CNN tennis table tennis rally point segmentation broadcast commentary"

Links: [{"title": "(PDF) Ball Hit Detection in Table Tennis Games Based on Audio Analysis", "url": "https://www.researchgate.net/publication/224649455_Ball_Hit_Detection_in_Table_Tennis_Games_Based_on_Audio_Analysis"}, {"title": "Spin Detection Using Racket Bounce Sounds in Table Tennis This research was funded by Sony AI.", "url": "https://arxiv.org/html/2409.11760v1"}, {"title": "(PDF) Spin Detection Using Racket Bounce Sounds in Table Tennis", "url": "https://www.researchgate.net/publication/384115010_Spin_Detection_Using_Racket_Bounce_Sounds_in_Table_Tennis"}, {"title": "Detection of Ball Spin Direction using Hitting Sound in Tennis | Request PDF", "url": "https://www.researchgate.net/publication/346826680_Detection_of_Ball_Spin_Direction_using_Hitting_Sound_in_Tennis"}, {"title": "(PDF) Detection of ball hits in a tennis game using audio and visual information", "url": "https://www.researchgate.net/publication/261119682_Detection_of_ball_hits_in_a_tennis_game_using_audio_and_visual_information"}, {"title": "Sound-Based Spin Estimation in Table Tennis: Dataset and Real-Time Classification Pipeline This work was funded by Sony AI", "url": "https://arxiv.org/html/2409.11760"}, {"title": "Spin Detection Using Racket Bounce Sounds in Table Tennis", "url": "https://arxiv.org/abs/2409.11760"}, {"title": "Sound-Based Spin Estimation in Table Tennis: Dataset and Real-Time Classification Pipeline", "url": "https://arxiv.org/pdf/2409.11760"}]

I found relevant research on audio onset detection for racket sports. Here are the key findings:

Audio Onset Detection in Racket Sports

The detection and classification of racket-ball bounce sounds falls within Sound Event Detection (SED), which detects the temporal onset and offset of different sound events and classifies them. SED offers potential for automatic annotation in sports, such as detecting highlights in baseball, golf, soccer, and improving annotation of tennis games.

Table Tennis Applications

In table tennis, the brief, high-frequency sounds produced during racket-ball impacts carry information about the racket type, the surface contacted, and whether spin was applied. Researchers address three key problems: precise bounce detection with millisecond-level temporal accuracy, classification of bounce surface, and spin detection, using a pipeline combining energy-based peak detection with CNNs trained on a dataset of 3,396 bounce samples.

CNN-Based Detection Methods

The first neural network approach for detecting tennis hits used a CNN taking MFCC and delta coefficients as inputs, achieving classification accuracy over 90% but with low onset accuracy due to 20 ms time frames. More recent approaches first detect high frequency peaks corresponding to ball bounces, then refine results using a CNN-based classifier to predict racket type and whether spin was applied.

The research demonstrates practical applications for automated sports analysis, including potential integration with broadcast commentary systems for real-time event detection and segmentation.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

7. user (2026-06-06T17:29:25.610Z)

Web search results for query: "few labels overfitting pseudo-label noise VLM teacher distillation video small dataset data efficiency vs weak supervision"

Links: [{"title": "A Review of Pseudo Labeling for Semi-Supervised ...", "url": "https://arxiv.org/pdf/2408.07221"}, {"title": "Remote Sensing Image Scene Classification with Noisy Label Distillation", "url": "https://www.mdpi.com/2072-4292/12/15/2376"}, {"title": "(PDF) Learning from Noisy Labels with Distillation", "url": "https://www.academia.edu/127989962/Learning_from_Noisy_Labels_with_Distillation"}, {"title": "Learning from Noisy Labels with Distillation Yuncheng Li Jianchao Yang", "url": "https://arxiv.org/pdf/1703.02391"}, {"title": "Ranking Distillation for Open-Ended Video Question Answering with Insufficient Labels", "url": "https://arxiv.org/html/2403.14430v1"}, {"title": "VIOLA: Towards Video In-Context Learning with Minimal Annotations", "url": "https://arxiv.org/pdf/2601.15549"}, {"title": "Knowledge Distillation and Student-Teacher Learning for Visual Intelligence: A Review and New Outlooks", "url": "https://arxiv.org/pdf/2004.05937"}, {"title": "Tiny is not small enough: High-quality, low-resource facial animation models through hybrid knowledge distillation", "url": "https://arxiv.org/pdf/2507.18352"}, {"title": "Pseudo-Label Correction for Instance-Dependent Noise Using Teacher-Student Framework", "url": "https://arxiv.org/html/2311.14237"}]

I've performed a web search on your query and found several relevant research results. Here's what the search returned:

The results cover several interconnected topics in your query:

Pseudo-labeling and Noise: The accumulation of false positive labels during pseudo labeling leads to incremental noise, thus limiting further improvements in performance, which is particularly problematic in limited label scenarios.

Video and Few-Shot Learning: Unlike text domain where models can leverage many-shot learning to statistically mitigate label noise, video inputs incur prohibitive token costs, confining video in-context learning to a strict few-shot regime where a single erroneous pseudo-label can induce significant negative transfer.

Teacher-Student Distillation with Noisy Labels: A distillation process can be used to learn from noisy datasets, assuming a small clean dataset and a large noisy dataset. Additionally, leveraging large numbers of noisy labels to augment small amounts of clean labels can improve the generalization and robustness of student networks.

Small Dataset Efficiency: Relying solely on a small labeled dataset to train efficient models is not sufficient to achieve high performance, and pseudo-labels can overcome the lack of labels.

Overfitting in Few-Label Settings: A common problem with models trained on few labeled examples is overfitting.

The search results highlight the tension between data efficiency and weak supervision through

pseudo-labeling in constrained video and small-dataset scenarios.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. assistant (2026-06-06T17:29:28.886Z)

I have strong coverage now. Let me run two more targeted searches: one on pose/court-geometry state machines for serve detection, and one to confirm the ShuttleSet license + TrackNet datasets for commercial use.

9. user (2026-06-06T17:29:37.877Z)

Web search results for query: "badminton serve detection player pose court homography rally start state machine self-supervised weak labels broadcast"

Links: [{"title": "Service and End of Rally Detection in Badminton Videos | Springer Nature Link", "url": "https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15"}, {"title": "(PDF) Towards Structured Analysis of Broadcast Badminton Videos", "url": "https://www.researchgate.net/publication/322075902_Towards_Structured_Analysis_of_Broadcast_Badminton_Videos"}, {"title": "Towards Structured Analysis of Broadcast Badminton Videos", "url": "https://arxiv.org/pdf/1712.08714"}, {"title": "Exploration of Player Behaviors from Broadcast Badminton ...", "url": "https://people.cs.nycu.edu.tw/~yushuen/data/BadmintonVis23.pdf"}, {"title": "Automated Service Height Fault Detection Using Computer Vision and Machine Learning for Badminton Matches", "url": "https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10747833/"}, {"title": "BST: Badminton Stroke-type Transformer for Skeleton-based Action Recognition in Racket Sports", "url": "https://arxiv.org/html/2502.21085v2"}, {"title": "ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis", "url": "https://arxiv.org/pdf/2306.04948"}, {"title": "Open Source Sports Video Analysis using Machine Learning - DEV Community", "url": "https://dev.to/stephan007/open-source-sports-video-analysis-using-machine-learning-2ag4"}]

Based on the web search results, here's what I found related to your query about badminton serve detection and rally analysis:

Service and Rally Detection

A technique has been presented to detect the start of service and end of rally in badminton videos, which utilizes court details and poses to accurately identify the service action from other badminton actions. The characteristics of service action are utilized to obtain features arranged in a 2D format for a convolutional neural network (CNN) to detect service and non-service.

Broadcast Video Analysis

An end-to-end framework has been proposed to automatically annotate badminton broadcast videos without relying on special camera setup or additional sensors. The task of analyzing unstructured broadcast videos is complicated by cameras having oblique or overhead views, where crucial aspects such as wrist and leg movements may not be visible in the same frame.

Player Pose and Court Homography

The perspective court view in video can be transformed to a standard top-view court coordinate system using homography to avoid the influence of depth perception. Data preprocessing involves producing player poses, positions, and shuttlecock trajectories from stroke clips.

Rally Structure

A rally is a sequence of strokes that start from a serve and ends when the shuttlecock falls on the court and a point is awarded to a player.

The search results cover several relevant topics from your query including serve detection, player pose analysis, court homography transformation, and rally structure analysis in badminton broadcast videos.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

10. user (2026-06-06T17:30:05.715Z)

Web search results for query: "ShuttleSet TrackNetV2 badminton dataset license CC non-commercial commercial use research only weights"

Links: [{"title": "MonoTrack: Shuttle trajectory reconstruction from monocular badminton video", "url": "https://arxiv.org/pdf/2204.01899"}, {"title": "Shuttlecock Trajectory Dataset - HackMD", "url": "https://hackmd.io/@TUIK/rJkRW54cU"}, {"title": "000 001 002 003 004 005 006 007 008 009 010 011 012 013 014 015 016 017 018 019", "url": "https://openreview.net/pdf/e360f0b2cb53ae8733ca7f6703925a362e9d94bd.pdf"}, {"title": "BST: Badminton Stroke-type Transformer for Skeleton-based Action Recognition in Racket Sports", "url": "https://arxiv.org/html/2502.21085v2"}, {"title": "TrackNetV2: Efficient Shuttlecock Tracking Network | IEEE Conference Publication | IEEE Xplore", "url": "https://ieeexplore.ieee.org/document/9302757/"}, {"title": "ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis", "url": "https://arxiv.org/pdf/2306.04948"}, {"title": "ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis | Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining", "url": "https://dl.acm.org/doi/10.1145/3580305.3599906"}, {"title": "[2306.04948] ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis", "url": "https://arxiv.org/abs/2306.04948"}, {"title": "(PDF) ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis", "url": "https://www.researchgate.net/publication/371413847_ShuttleSet_A_Human-Annotated_Stroke-Level_Singles_Dataset_for_Badminton_Tactical_Analysis"}]

Let me search for more specific information about the license terms for these datasets.

Links: [{"title": "UCI Machine Learning Repository", "url": "https://archive.ics.uci.edu/dataset/148/statlog+shuttle"}, {"title": "Research data how to license", "url": "https://www.openaire.eu/research-data-how-to-license/"}, {"title": "How do I license my research data?", "url": "https://www.openaire.eu/how-do-i-license-my-research-data"}, {"title": "How to License Research Data | DCC", "url": "https://www.dcc.ac.uk/guidance/how-guides/license-research-data"}, {"title": "The Esethu Framework: Reimagining Sustainable Dataset Governance and Curation for Low-Resource Languages", "url": "https://arxiv.org/pdf/2502.15916"}, {"title": "Common license types for datasets - Documentation", "url": "https://docs.data.world/en/214274-common-license-types-for-datasets.html"}, {"title": "Common license types for datasets and what they mean - Get Help", "url": "https://libanswers.ucalgary.ca/faq/200582"}, {"title": "Licenses ? Research Data Management", "url": "https://rdm.mpg.de/after-research/licenses/"}]

Let me search more specifically for the ShuttleSet and TrackNetV2 project pages or repositories.

Links: [{"title": "GitHub - ChgygLin/TrackNetV2-pytorch: A Pytorch implementation of TrackNetV2 from Tensorflow (ncnn c++ inference) .", "url": "https://github.com/ChgygLin/TrackNetV2-pytorch"}, {"title": "GitHub - AndrewDettor/TrackNet-Pickleball: Deep Learning model to track a pickleball's location in a video of a match .", "url": "https://github.com/AndrewDettor/TrackNet-Pickleball"}, {"title": "GitHub - SamuReyes/TrackNetV2-padel: Improvement of TrackNetV2 and transfer learning for ball segmentation in padel videos", "url": "https://github.com/SamuReyes/TrackNetV2-padel"}, {"title": "GitHub - mareksubocz/TrackNet: :flashlight: Pytorch implementation of TrackNet .", "url": "https://github.com/mareksubocz/TrackNet"}, {"title": "GitHub - TrackNetV4/TrackNetV4 .", "url": "https://github.com/TrackNetV4/TrackNetV4"}, {"title": "GitHub - weekenddeeplearning/TrackNet: Heatmap based high speed tiny sport objects tracking based on TrackNet .", "url": "https://github.com/weekenddeeplearning/TrackNet"}, {"title": "TrackNetV4: Enhancing", "url": "https://github.com/TrackNetV4/TrackNetV4"}]

Fast Sports Object Tracking with Motion Attention

Maps", "url": "https://arxiv.org/pdf/2409.14543"}, {"title": "GitHub - Chang-Chia-Chi/TrackNet-Badminton-Tracking-tensorflow2: TrackNet for badminton tracking using tensorflow2 . GitHub", "url": "https://github.com/Chang-Chia-Chi/TrackNet-Badminton-Tracking-tensorflow2"}, {"title": "GitHub - yastrebkv/TrackNet: Unofficial PyTorch implementation of TrackNet . GitHub", "url": "https://github.com/yastrebkv/TrackNet"}]

Let me search for the official TrackNetV2 and ShuttleSet project pages with more specific license information.

Links: [{"title": "Licenses List - Creative Commons", "url": "https://creativecommons.org/licenses/list.en"}, {"title": "Sharing Openly, Sharing Globally - Creative Commons", "url": "https://creativecommons.org/cc-licenses/"}, {"title": "Creative Commons Licenses and Definitions - Prints & Photographs: Copyright and Rights and Restrictions Information - Research Guides at Library of Congress", "url": "https://guides.loc.gov/p-and-p-rights-and-restrictions/creative-commons-definitions"}, {"title": "3.3 License Types - Creative Commons", "url": "https://creativecommons.org/course/cc-cert-edu/unit-3-anatomy-of-a-cc-license/3-3-license-types/"}, {"title": "Creative Commons Licenses - Copyright - LibGuides at Utah State University", "url": "https://libguides.usu.edu/authorsrights/creativecommons"}, {"title": "Deed - Attribution-NonCommercial 4.0 International - Creative Commons", "url": "https://creativecommons.org/licenses/by-nc/4.0/deed.en"}, {"title": "Creative Commons Licenses - Creative Commons", "url": "https://creativecommons.org/share-your-work/use-remix/cc-licenses/"}, {"title": "About CC Licenses - Creative Commons", "url": "https://creativecommons.org/share-your-work/cclicenses/"}, {"title": "Legal Code - Attribution-NonCommercial-ShareAlike 4.0 International - Creative Commons", "url": "https://creativecommons.org/licenses/by-nc-sa/4.0/legalcode.en"}, {"title": "Creative Commons Licenses - Open Educational Resources (OER)* - LSC-CyFair Library Guides at Lone Star College-CyFair", "url": "https://cflibguides.lonestar.edu/oer/licenses"}]

Let me try one more specific search for the actual dataset documentation or project pages.

Links: [{"title": "ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis | Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining", "url": "https://dl.acm.org/doi/10.1145/3580305.3599906"}, {"title": "ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis | Request PDF", "url": "https://www.researchgate.net/publication/372910678_ShuttleSet_A_Human-Annotated_Stroke-Level_Singles_Dataset_for_Badminton_Tactical_Analysis"}, {"title": "ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis - ADS", "url": "https://ui.adsabs.harvard.edu/abs/2023arXiv230604948W/abstract"}, {"title": "Bridging the Gap: Doubles Badminton Analysis with Singles-Trained Models", "url": "https://arxiv.org/pdf/2508.13507"}, {"title": "ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis", "url": "https://arxiv.org/pdf/2306.04948"}, {"title": "GitHub - wywyWang/CoachAI-Projects: Official research projects of badminton CoachAI", "url": "https://github.com/wywyWang/CoachAI-Projects"}, {"title": "GitHub - muhammadyasin79/Badminton_Analytics_Project: Real-time badminton analytics using YOLOv8?Pose - skeleton overlay, player stats and trajectory visualisation with OpenCV & MoviePy.", "url": "https://github.com/muhammadyasin79/Badminton_Analytics_Project"}, {"title": "GitHub - ToanNguyenKhanh/Badminton-Analysis: This project utilizes computer vision techniques to detect players and shuttlecocks in badminton games. . GitHub", "url": "https://github.com/ToanNguyenKhanh/Badminton-Analysis"}]

Based on my web search results, I found information about ShuttleSet and TrackNetV2 badminton datasets. Here's what I discovered:

ShuttleSet Dataset

The ShuttleSet dataset and corresponding benchmarks are available at <https://github.com/wywyWang/CoachAI-Projects>. However, the search results did not reveal specific license information (such as CC-BY-NC or other Creative Commons licenses) in the publicly available documentation I could access.

TrackNetV2 Dataset

The TrackNetV2 dataset resources can be found at the project's official sources, with the TrackNet project at <https://nol.cs.nctu.edu.tw:234/open-source/TrackNet> and TrackNetV2 at <https://nol.cs.nctu.edu.tw:234/open-source/TrackNetv2>.

General Notes on the Datasets

ShuttleSet is the largest publicly-available badminton singles dataset with annotated stroke-level records, containing 104 sets, 3,685 rallies, and 36,492 strokes in 44 matches between 2018 and 2021.

****Important Note:**** The web search results did not contain explicit license statements (such as CC-BY-NC, CC-BY-NC-SA, or other Creative Commons licenses) for either dataset. To find the specific license terms for non-commercial vs. commercial use and restrictions on research-only access, I recommend:

1. Checking the GitHub repositories directly (<https://github.com/wywyWang/CoachAI-Projects> for ShuttleSet)
2. Looking for LICENSE files or README documentation in those repositories
3. Contacting the dataset creators directly for explicit license terms

The search results focused on the paper descriptions and general information rather than the licensing terms you're inquiring about.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

11. assistant (2026-06-06T17:30:10.360Z)

I have enough strong material. Let me verify the ShuttleSet license directly from its GitHub repo, since commercial viability is a hard constraint for the user.

12. user (2026-06-06T17:30:16.386Z)

License Information

Based on the GitHub repository content, the CoachAI-Projects (which includes ShuttleSet) are released under the ****MIT license****, as indicated by the repository's license file reference.

MIT is a permissive, commercial-friendly open source license that allows:

- Commercial use
- Modification
- Distribution
- Private use

The repository footer states: "[MIT license](#MIT-1-ov-file)" confirming this permissive licensing approach.

Since MIT is one of the most commercially permissive licenses available, both the ShuttleSet dataset and CoachAI projects can be used for commercial purposes, provided the license terms are respected.

13. assistant (2026-06-06T17:30:20.744Z)

I have all the material I need. The repo code is MIT, though the underlying broadcast video data is a separate question. I have strong coverage across all angles: the contrarian distillation paper, weak supervision baselines, audio hit detection, pose/court state machines, the ShuttleSet dataset/license, and the few-labels overfitting evidence. Let me return the ranked results.

14. user (2026-06-06T17:30:45.887Z)

Structured output provided successfully

15. assistant (2026-06-06T17:30:48.143Z)

Returned 6 ranked results covering all angles of the contrarian distillation-vs-weak research question.